

AI System Specification: Medical Imaging Assistant

Assignment 1 | Topic: PEAS & Task Environment Analysis

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Task 1: PEAS Specification

Performance Measures: Accuracy in detecting early-stage disease (Sensitivity); speed of prioritizing life-threatening cases; rate of false positives that cause clinician fatigue; and transparency of results.

Environment: Digital hospital radiology units, integrated with cloud-based PACS systems and clinical data pipelines.

Actuators: Graphical User Interface (GUI) on clinician terminals, push notifications for emergency triage, and report auto-population tools.

Sensors: High-resolution DICOM image metadata, patient history records from EHR, and direct feedback from radiologist manual overrides.

Task 2: Environment Classification

- **Partially Observable:** The agent cannot observe the patient's internal biology directly; it only sees the 2D/3D visual slices provided.
- **Multi-agent:** The AI works in an ecosystem with human doctors and other hospital management software agents.
- **Stochastic:** The same visual pattern could result in different biological outcomes, making the environment probabilistic.
- **Sequential:** Current prioritization impacts the time available for future patient scans and diagnostic workflows.
- **Dynamic:** Hospital emergencies and new scan arrivals change the "state" of the worklist constantly.
- **Continuous:** Pixel values in medical images and the timing of incoming cases are not restricted to discrete steps.

Task 3: Critical Analysis

The Core Technical Challenge

The **Stochastic** nature of medical data is the most significant hurdle. Unlike logic-based systems, medical AI must handle high uncertainty. A "humanized" design requires the AI to communicate this uncertainty to the doctor—not just providing a label, but a confidence interval that respects the gravity of a medical decision

Utility Function & Trade-offs

Proposed Function: $U = 0.7 * (\text{Sensitivity}) - 0.3 * (\text{False_Alarm_Cost})$

This balances finding every potential illness against the risk of "crying wolf." If the weight of sensitivity were doubled, the system would become hyper-cautious, flagging almost every patient for review. While this minimizes missed diagnoses, it would likely lead to "alert fatigue" where human doctors start ignoring the AI entirely because its signals become too noisy.

Task 4: Structured Representation (JSON)

```
{
  "system_name": "MediScan Assistant (Medical Imaging Triage)",
  "peas": {
    "performance_measures": [
      "Diagnostic Sensitivity (Early Detection Rate)",
      "Triage Prioritization Speed",
      "Clinician False-Alarm Rate",
      "Algorithm Explainability Score"
    ],
    "environment": "Digital Radiology Department & Hospital PACS Workflow",
    "actuators": [
      "Diagnostic UI Dashboard",
      "Emergency Push Notifications",
      "Automated Report Annotator",
      "Specialist Routing Module"
    ],
    "sensors": [
      "High-Resolution DICOM Image Stream",
      "Electronic Health Records (EHR) Metadata",
      "Clinician Feedback Interface",
      "Hospital Info System (HIS) API"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "The AI only sees digital images and metadata, missing the patient's full physical and genetic context."
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "The system operates alongside human radiologists and automated hospital scheduling agents."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Medical symptoms in images do not map to outcomes with 100% certainty; the environment is probabilistic."
    },
    "episodic_sequential": {
      "choice": "Sequential",
      "justification": "Initial triage decisions dictate the downstream availability of specialists and patient care paths."
    },
    "static_dynamic": {
      "choice": "Dynamic",
      "justification": "The environment changes as new emergency cases arrive and patient statuses update in real-time."
    }
  }
}
```

```
    },  
    "discrete_continuous": {  
      "choice": "Continuous",  
      "justification": "Image pixel intensities and the arrival times of medical  
cases occur on a continuous scale."  
    }  
  },  
  "utility_function": "U = 0.7*Sensitivity - 0.3*False_Alarm_Cost"  
}
```

Generated for Academic Submission - AI System Design Practice